



151-0854-00L

Autonomous Mobile Robots

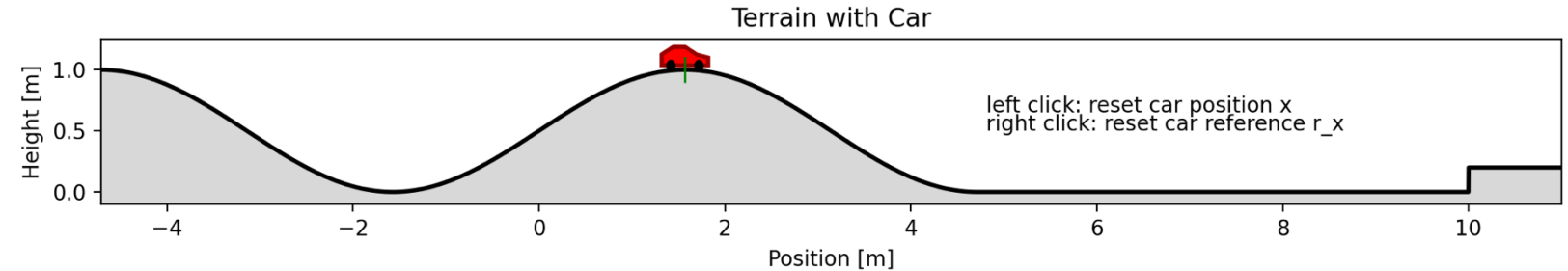
E10: Feedback Control

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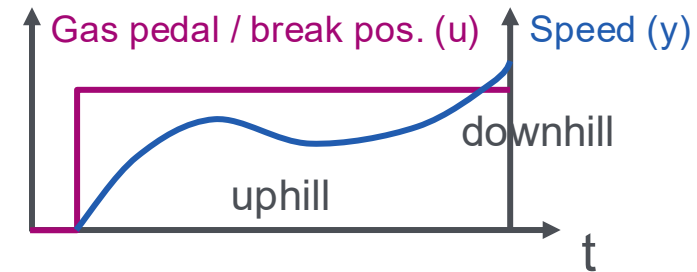
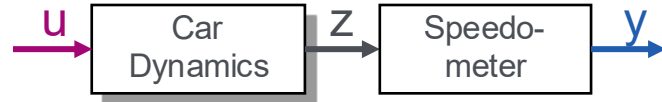


Feedback Control of Dynamical Systems

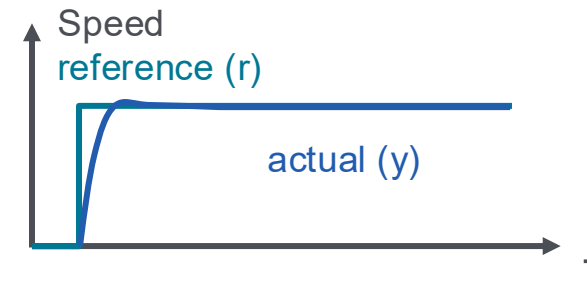
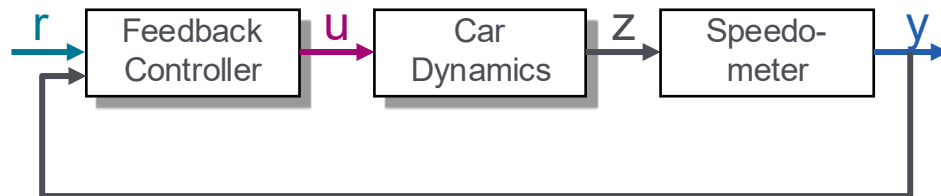
Ex: Cruise control of a car



Open-loop response



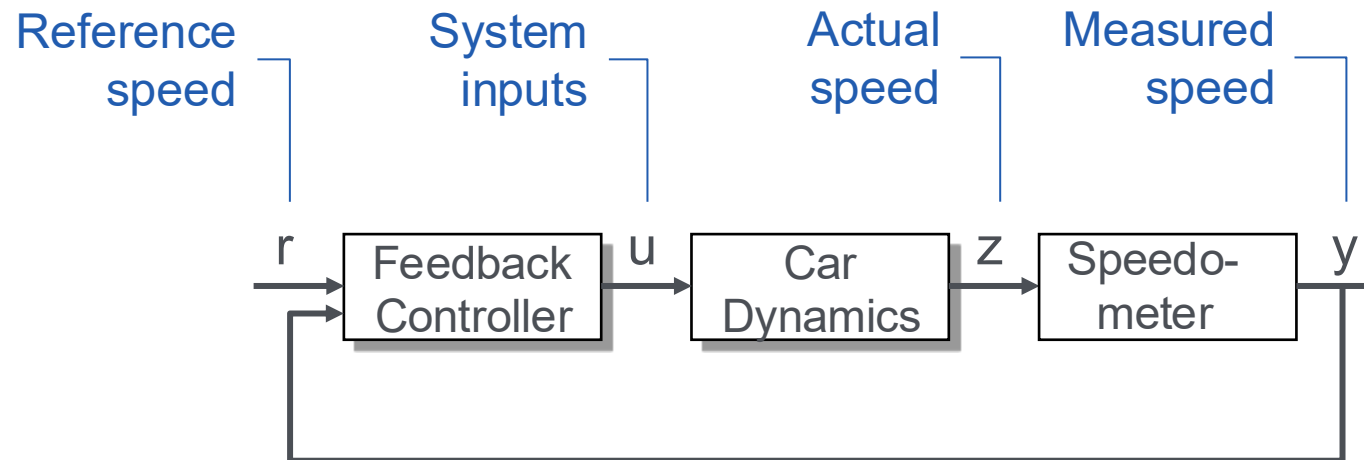
Closed-loop response



Single-Input-Single-Output (SISO) Systems

Example: cruise control

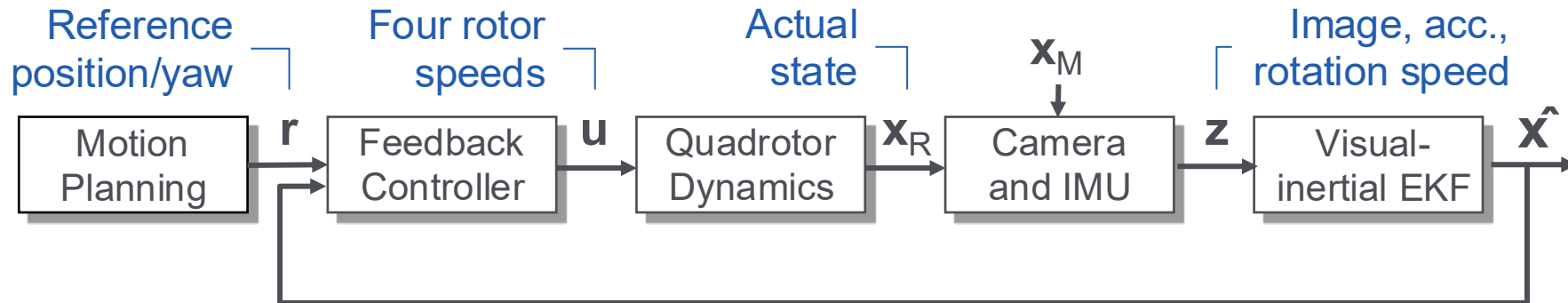
- Reference: r : the desired speed
- System input: u : the gas pedal position ($u > 0$) / break position ($u < 0$)
- System output: z : the speed of the car



Most Systems Are Multiple-Input-Multiple-Output (MIMO)

Example: quadrotor

- Reference: $\mathbf{r}$: typically: position/yaw (maybe also speed?) trajectory
- System inputs: $\mathbf{u}$: four rotor speeds
- Internal states $\mathbf{x}$: position, orientation, speed, rot. speed, (IMU biases)
- System outputs: $\hat{\mathbf{x}}$: typically: pos., orientation, speed, rot. speed (IMU biases)

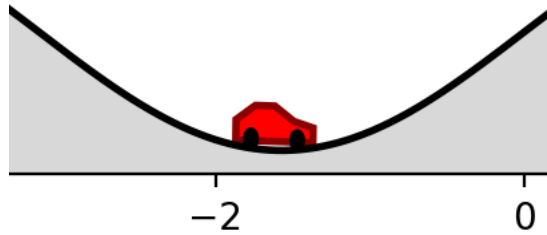


What could a feedback control law look like?

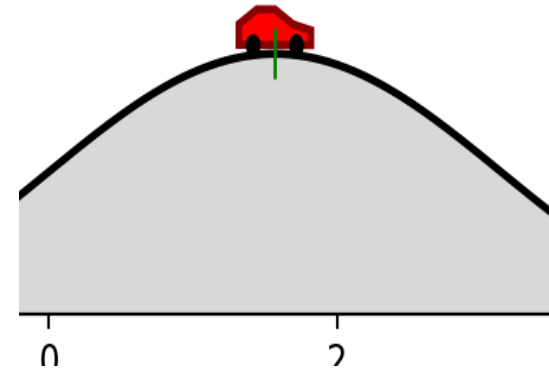
- This is not obvious at all... We will see
- how we can still do SISO control with some trickery,
- or we can do actual MIMO control (e.g. model-predictive).

Stability of Dynamical Systems

Ex: Position of the car (open-loop = no feedback control) → Task 1.i



stable equilibrium
(will return to bottom after disturbance)



unstable equilibrium
(won't return to top after disturbance)

Principled way quantify stability?

Stability of Dynamical Systems

Principled way quantify stability? → **Task 1.i**

$$\begin{aligned} \lim_{t \rightarrow \infty} \mathbf{x}(t) &= \mathbf{0} \\ \lim_{k \rightarrow \infty} \mathbf{x}_k &= \mathbf{0} \end{aligned} \quad \begin{array}{l} \text{Stability Criterion} \\ (\mathbf{x} = \mathbf{0}, \mathbf{u} = \mathbf{0} \\ \text{is equilibrium}) \end{array}$$

Linear time-invariant (LTI) system in continuous time

Linear time-invariant (LTI) system in discrete time

$$\dot{\mathbf{x}}(t) = \mathbf{F}_c \mathbf{x}(t) + \mathbf{G}_c \mathbf{u}(t).$$

$$\dot{\boldsymbol{\xi}}(t) = \text{diag}(\boldsymbol{\lambda}) \boldsymbol{\xi}(t),$$

$$\xi_i(t) = \xi_i(0) \exp \lambda_i t.$$

Eigendecomposition
 $\mathbf{F} = \mathbf{Q} \text{diag}(\boldsymbol{\lambda}) \mathbf{Q}^{-1}$
 $\boldsymbol{\xi}(t) = \mathbf{Q}^{-1} \mathbf{x}(t)$

Integrate/solve ODE
 scalar difference eq.

$$\mathbf{x}_k = \mathbf{F} \mathbf{x}_{k-1} + \mathbf{G} \mathbf{u}_k.$$

$$\boldsymbol{\xi}_k = \text{diag}(\boldsymbol{\lambda}) \boldsymbol{\xi}_{k-1},$$

$$\xi_{i,k} = \xi_0 \prod_{\kappa=1}^k \lambda_i = \xi_0 \lambda_i^k.$$

$$\lambda_i = a_i + i\omega_i$$

$$a_i < 0, \forall i$$

$$\lim_{t \rightarrow \infty} \mathbf{x}(t) = \lim_{t \rightarrow \infty} \mathbf{Q} \boldsymbol{\xi}(t) = \mathbf{0}$$

Stability Criteria
 ($\mathbf{x} = \mathbf{0}$ is equilibrium)

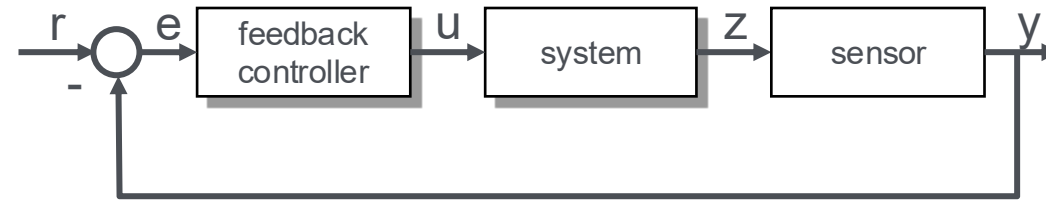
$$\lambda_i = r_i (\cos \varphi_i + i \sin(\varphi_i)), r_i = \sqrt{a_i^2 + b_i^2}$$

$$r_i < 1, \forall i$$

$$\lim_{k \rightarrow \infty} \mathbf{x}_k = \lim_{k \rightarrow \infty} \mathbf{Q} \boldsymbol{\xi}_k = \mathbf{0}$$

Proportional-Integral-Differential (PID) Control

→ Task 1.ii



$$u(t) = k_p e(t) + k_i \int_{t_0}^t e(\tau) d\tau + k_d \frac{de(t)}{dt}$$

reduces current error

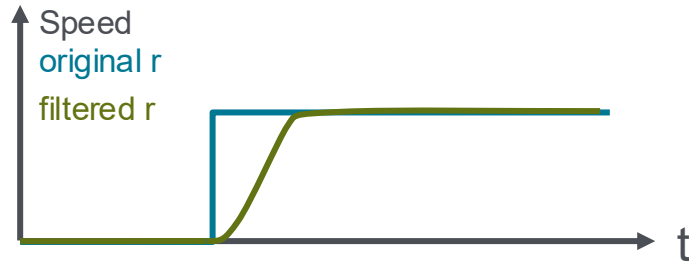
reduces long-term offset

considers error trend,
reduces overshoot

PID Control: Popular Extensions

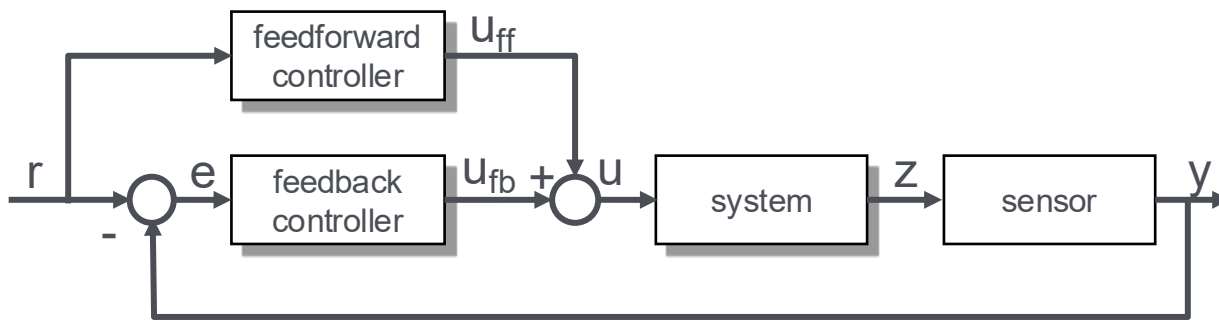
→ Task 1.iii

1. **Reference filtering**: respect physical limits already in the reference. E.g.



2. **Anti-Reset-Windup**: stops integrating the error for the I-part, when u is at its physical limit.

3. **Feed-forward controller**: reduce “work” for the feed-back controller.
E.g. nominal gas pedal positions resulting in a certain speed.



General Optimal Control Problem Formulation

→ Task 2

We will consider discrete-time **performance objectives** or **cost functions**

$$J_{0 \rightarrow N}(\mathbf{x}_0, \mathbf{u}_0, \dots, \mathbf{u}_{N-1}) = \boxed{p(\mathbf{x}_N)} + \boxed{\sum_{k=0}^{N-1} q(\mathbf{x}_k, \mathbf{u}_k)}$$

Terminal cost **Stage cost**

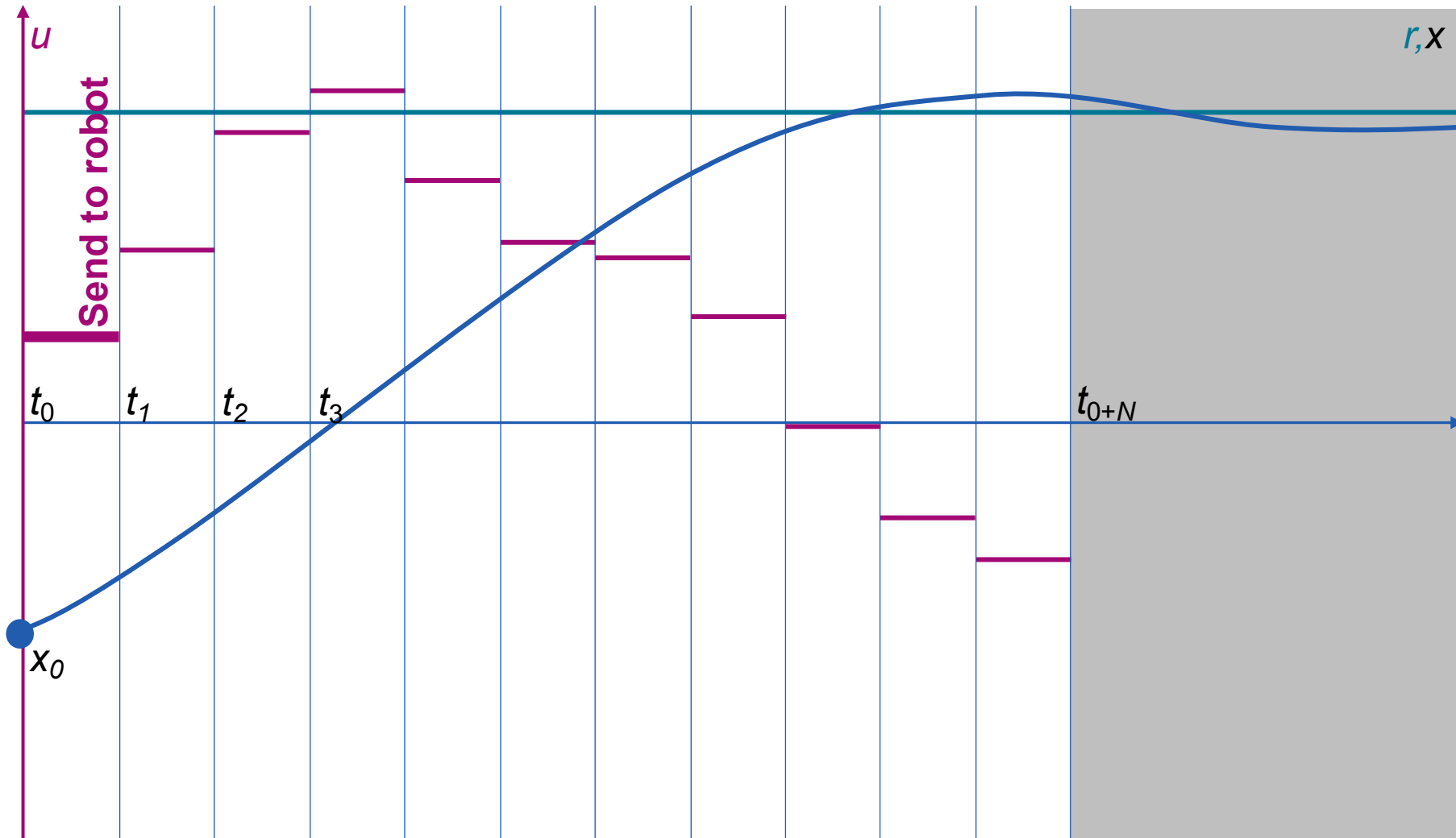
that we minimise w.r.t. the controls:

$$J_{0 \rightarrow N}^*(\mathbf{x}_0) = \min_{\mathbf{u}_{0 \rightarrow N}} J_{0 \rightarrow N}(\mathbf{x}_0, \mathbf{u}_0, \dots, \mathbf{u}_{N-1})$$

s.t. $\mathbf{x}_{k+1} = \mathbf{f}(\mathbf{x}_k, \mathbf{u}_k), k = 0, \dots, N-1,$ **System dynamics**
 $\mathbf{g}(\mathbf{x}_k, \mathbf{u}_k) \leq \mathbf{0}, k = 0, \dots, N-1,$ **State/input constraints**
 $\mathbf{x}_N \in \mathcal{X}_f,$ **Terminal state constr.**
 $\mathbf{x}_0 = \mathbf{x}(0).$ **Initial condition**

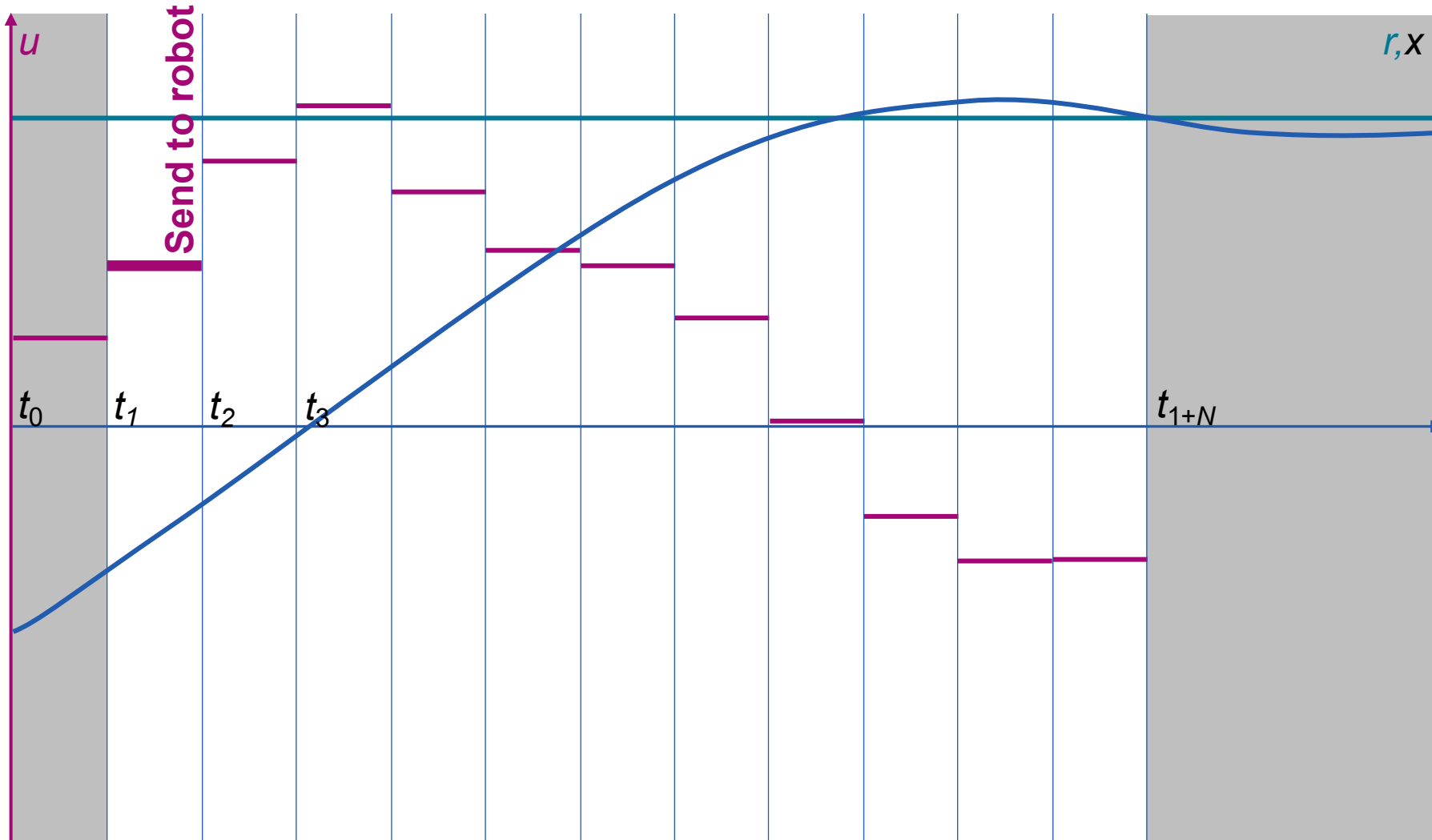
MPC: Application as Receding Horizon Controller

→ Task 2



MPC: Application as Receding Horizon Controller

→ Task 2



Linear-Quadratic Optimal Control with Constraints

→ Task 2.ii

We can consider a **quadratic cost**

$$J_{0 \rightarrow N}(\mathbf{x}_0, \mathbf{u}_0, \dots, \mathbf{u}_{N-1}) = \mathbf{x}_N^T \mathbf{P} \mathbf{x}_N + \sum_{k=0}^{N-1} \mathbf{x}_k^T \mathbf{Q} \mathbf{x}_k + \mathbf{u}_k^T \mathbf{R} \mathbf{u}_k$$

But s.t. **linear system dynamics** and **polyhedral input and state constraints**:

$$\begin{aligned} J_{0 \rightarrow N}^*(\mathbf{x}_0) = & \min_{\mathbf{u}_{0 \rightarrow N}} J_{0 \rightarrow N}(\mathbf{x}_0, \mathbf{u}_0, \dots, \mathbf{u}_{N-1}) \\ \text{s.t. } & \mathbf{x}_{k+1} = \mathbf{F} \mathbf{x}_k + \mathbf{G} \mathbf{u}_k, \quad k = 0, \dots, N-1, \\ & \mathbf{x}_k \in \mathcal{X}, \quad \mathbf{u}_k \in \mathcal{U}, \quad k = 0, \dots, N-1, \\ & \mathbf{x}_N \in \mathcal{X}_f, \\ & \mathbf{x}_0 = \mathbf{x}(0). \end{aligned}$$

Using the same trick of substitution in the batch solution approach, we now don't obtain an unconstrained square cost in the control sequence, but instead a respective **Quadratic Program** (QP). This can even be turned into an explicit control law $\mathbf{u}_0^* = \mathbf{u}^*(\mathbf{x}(0))$ as a **Multiparametric Program** (not explained here).